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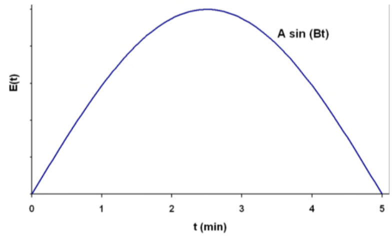
HW 11

3. (50 points) A first order reaction

$$A \xrightarrow{k} B$$

with k = 2 min^{-1} is taking place in a real reactor with the following RTD function



1)

For $0 < t < 5$ min; $E(t) = A \sin (Bt)$
$t > 5$ min; $E(t) = 0$

(a) What is the mean residence time?
(b) What is the conversion predicted by the segregation model?
(c) If there is no reaction taking place, what is the probability that a molecule entering the reactor spends a time between 5/3 min and 5/2 min inside the reactor?

A) Mean RT from graph $= \frac{5-0}{2} = \boxed{2.5 \text{ min}}$

B) $\bar{X} = \int_0^5 X(t)\, E(t)\, dt \qquad X(t) = 1 - e^{-kt} \qquad E(t) = A\sin(Bt)$

$\bar{X} = \int_0^5 (1-e^{2t}) \frac{\pi}{10} \sin\left(\frac{\pi t}{5}\right) dt =$

C) probability $= \int_{5/3}^{5/2} \frac{\pi}{10} \sin\left(\frac{\pi}{5}t\right) dt = \frac{\pi}{10}\frac{5}{\pi}\int_{\pi/3}^{\pi/2} \sin(u)\, du \qquad u = \frac{\pi x}{5}$

$\frac{\pi}{10}\frac{5}{\pi}\left[-\cos(u)\right]_{\pi/3}^{\pi/2} = \frac{1}{2}\left[-\cos\frac{\pi}{2} + \cos\frac{\pi}{3}\right] = \frac{1}{2}\cdot\frac{1}{2} = \boxed{\frac{1}{4}}$

2) FDK